Problem 1 Temperature Status

Read a temperature in Celsius. Print “Freezing” if it is below 0; otherwise print “Not freezing”. Test both branches.

1 Pseudocode

Initialize variable with temperature
Cout and Cin for output and input
if and else statement, comparing variable to number value

2 C++ Source File

The .cpp file has been uploaded separately.

3 Output Screenshot

```
georgy@Georgy:~$ cd EECE2140/Week2/Classwork
georgy@Georgy:~/EECE2140/Week2/Classwork$ g++ InClassWork03_Problem01_Rached.cpp -o p1
georgy@Georgy:~/EECE2140/Week2/Classwork$ ./p1
Enter temperature: -3
Freezing
georgy@Georgy:~/EECE2140/Week2/Classwork$ ./p1
Enter temperature: 5
Not Freezing
georgy@Georgy:~/EECE2140/Week2/Classwork$
```

Problem 2 Even or Odd Device ID

Read a positive device ID. Use % to determine whether it is even or odd, and display the result.

1 Pseudocode

Input int id
If id%2 == 0
Even
Else odd

2 C++ Source File

The .cpp file has been uploaded separately.

3 Output Screenshot

```
georgy@Georgy:~/EECE2140/Week2/Classwork$ g++ InClassWork03_Problem02_Rached.cpp -o p2
georgy@Georgy:~/EECE2140/Week2/Classwork$ ./p2
Enter device ID: 27
Odd device ID
georgy@Georgy:~/EECE2140/Week2/Classwork$ ./p2
Enter device ID: 28
Even device ID
```

Problem 3 Voltage Warning

Read a measured voltage. Print “Low” when it is below 4.5 V, “High” when it is above 5.5 V, and “Normal” otherwise. Test each case.

1 Pseudocode

Initialize variable with voltage
Cout and Cin for output and input
Use if statements to compare voltage reading and output with cout

2 C++ Source File

The .cpp file has been uploaded separately.

3 Output Screenshot

```
georgy@Georgy:~/EECE2140/Week2/Classwork$ g++ InClassWork03_Problem03_Rached.cpp -o p3
georgy@Georgy:~/EECE2140/Week2/Classwork$ ./p3
Enter voltage: 5.2
Normal
georgy@Georgy:~/EECE2140/Week2/Classwork$ ./p3
Enter voltage: 1
Low
georgy@Georgy:~/EECE2140/Week2/Classwork$ ./p3
Enter voltage: 1000
High
```

Problem 4 Package Weight Category

Read a nonnegative weight in kilograms. Use an if / else if / else chain: up to and including 2 kg is Small; above 2 through 10 kg is Medium; above 10 kg is Large.

1 Pseudocode

Input int weight
If <=2 print small
Elif <= 10 print medium
Else print large

2 C++ Source File

The .cpp file has been uploaded separately.

3 Output Screenshot

```
georgy@Georgy:~/EECE2140/Week2/Classwork$ g++ InClassWork03_Problem04_Rached.cpp -o p4
georgy@Georgy:~/EECE2140/Week2/Classwork$ ./p4
Enter weight (kg): 10
Medium
georgy@Georgy:~/EECE2140/Week2/Classwork$ ./p4
Enter weight (kg): 1
Small
georgy@Georgy:~/EECE2140/Week2/Classwork$ ./p4
Enter weight (kg): 100
Large
```

Problem 5 Pump Safety Decision

Read the water level (0 to 100) and a maintenance flag (0 or 1). First check whether the pump is in maintenance. If it is not, use a nested if / else to start the pump when the level is below 25; otherwise keep it off.

1 Pseudocode

Initialize flag and level variable
Get value from cout and cin
Use nested if loop to confirm maintenance flag
If 0, measure water level with if and cout

2 C++ Source File

The .cpp file has been uploaded separately.

3 Output Screenshot

```
georgy@Georgy:~/EECE2140/Week2/Classwork$ ./p5
Enter water level: 18
Maintenance flag (0 or 1): 0
Pump on
georgy@Georgy:~/EECE2140/Week2/Classwork$ ./p5
Enter water level: 18
Maintenance flag (0 or 1): 1
Pump on maintenance
georgy@Georgy:~/EECE2140/Week2/Classwork$ ./p5
Enter water level: 28
Maintenance flag (0 or 1): 0
Pump off
```

Problem 6 Sensor Sample Numbers

Read a positive count n. Use a for loop to print the sample numbers from 1 through n on one line.

1 Pseudocode

Input int count
For i = 1 to count
Print I
I++

2 C++ Source File

The .cpp file has been uploaded separately.

3 Output Screenshot

```
georgy@Georgy:~/EECE2140/Week2/Classwork$ g++ InClassWork03_Problem06_Rached.cpp -o p6
georgy@Georgy:~/EECE2140/Week2/Classwork$ ./p6
Enter sample count: 5
1 2 3 4 5
```

Problem 7 Total Energy Over Days

Read the number of days n. Use a for loop to read one whole-number energy value (kWh) for each day, then print the total. Assume n is positive.

1 Pseudocode

Initialize days kwh and total
Get days total with cout

Use for loop, incrementing by 1, stops when i less than or equal to days
In for loop, ask for kwh
Add kwh to total variable
Cout total after for loop done

2 C++ Source File

The .cpp file has been uploaded separately.

3 Output Screenshot

```
georgy@Georgy:~/EECE2140/Week2/Classwork$ g++ InClassWork03_Problem07_Rached.cpp -o p7
georgy@Georgy:~/EECE2140/Week2/Classwork$ ./p7
Enter number of days: 3
Day 1 kWh: 4
Day 2 kWh: 6
Day 3 kWh: 5
Total: 15 kWh
```

Problem 8 Even-Number Countdown

Read a nonnegative integer n. Use a for loop to print all even numbers from n down to 0. Do not assume n itself is even.

1 Pseudocode

Input int count
For i = 1 to count
Print I
I++

2 C++ Source File

The .cpp file has been uploaded separately.

3 Output Screenshot

```
georgy@Georgy:~/EECE2140/Week2/Classwork$ g++ InClassWork03_Problem08_Rached.cpp -o p8
georgy@Georgy:~/EECE2140/Week2/Classwork$ ./p8
Enter n: 7
6 4 2 0
```

Problem 9 Resistor Power Table

Read a positive integer maximum voltage V and a positive resistance R. For each whole-number voltage from 1 to V, calculate power P = voltage * voltage / R using double arithmetic. Print each voltage and its power to two decimal places.

1 Pseudocode

Initialize 5 variables, maxVoltage, resistance, power, numberProblem and voltage
Set numberproblem and voltage to 1
Cout and Cin for max voltage and resistance
if maxV greater than 0, Make for loop that increments i by 1 until i less than or = to maxV
Do power calculation in for loop, display with cout, increment voltage
else Max V less then 0, say voltage below 0

2 C++ Source File

The .cpp file has been uploaded separately.

3 Output Screenshot

```
georgy@Georgy:~/EECE2140/Week2/Classwork$ g++ InClassWork03_Problem09_Rached.cpp -o p9
georgy@Georgy:~/EECE2140/Week2/Classwork$ ./p9
Enter maximum voltage: 3
Enter resistance: 2
1 V: 0.50 W
2 V: 2.00 W
3 V: 4.50 W
```

Problem 10 Repeated Growth

Read an initial integer population and a number of days. Each day the population doubles. Use a for loop to print the population after each day. Assume the inputs are small enough to fit in int.

1 Pseudocode

Input int pop
Input int days
For I = 1 to days
Pop x 2
Print i: pop
i++

2 C++ Source File

The .cpp file has been uploaded separately.

3 Output Screenshot

```
georgy@Georgy:~/EECE2140/Week2/Classwork$ g++ InClassWork03_Problem10_Rached.cpp -o p10
georgy@Georgy:~/EECE2140/Week2/Classwork$ ./p10
Enter initial population: 3
Enter days: 3
Day 1: 6
Day 2: 12
Day 3: 24
```

Problem 11 Battery Discharge

Read a starting battery percentage from 0 to 100. Use a while loop to subtract 7 percentage points per minute until the level reaches 0. Never print a negative level. Print the level after each minute.

1 Pseudocode

Initialize charge and minute
Cout and cin for battery percentage
Make while loop for charge>0
Subtract 7 from charge each time in loop, display charge, increment minute
Add if statement to print 0 if percentage becomes less than 0 to avoid negatives

2 C++ Source File

The .cpp file has been uploaded separately.

3 Output Screenshot

```
georgy@Georgy:~/EECE2140/Week2/Classwork$ g++ InClassWork03_Problem11_Rached.cpp -o p11
georgy@Georgy:~/EECE2140/Week2/Classwork$ ./p11
Enter battery percentage: 16
Minute 1: 9%
Minute 2: 2%
Minute 3: 0%
```

Problem 12 Valid Speed Input

Keep asking for a speed until the user enters a value from 0 through 120 inclusive. Then print the accepted value. Use a while loop.

1 Pseudocode

Input int speed
While speed not 0 – 120 Input int speed again
Print accepted: speed

2 C++ Source File

The .cpp file has been uploaded separately.

3 Output Screenshot

```
georgy@Georgy:~/EECE2140/Week2/Classwork$ ./p12
Enter speed (0-120): -2
Enter speed (0-120): 130
Enter speed (0-120): 55
Accepted: 55
```

Problem 13 Threshold Reading Count

Read sensor values one at a time until the user enters -1. Count the readings that are at least 70. The sentinel -1 must not count as a reading.

1 Pseudocode

Initialize tracker and reader
Cout and Cin for reading value
Make while look that continues until reading is -1
Cout for reading everytime in while loop
In while loop, use if statement to see if reading>70 and add to reading
Cout total at the end, outside of while loop

2 C++ Source File

The .cpp file has been uploaded separately.

3 Output Screenshot

```
georgy@Georgy:~/EECE2140/Week2/Classwork$ ./p13
Enter reading (-1 to stop): 65
Enter reading (-1 to stop): 78
Enter reading (-1 to stop): 53
Enter reading (-1 to stop): -1
Reading at least 70: 1
```

Problem 14 Number of Digits

Read a positive integer. Repeatedly divide it by 10 using a while loop to count its decimal digits. Print the count; do not convert the number to a string.

1 Pseudocode

Input int n
While n/10 != 0
Count++
Print count

2 C++ Source File

The .cpp file has been uploaded separately.

3 Output Screenshot

```
georgy@Georgy:~/EECE2140/Week2/Classwork$ g++ InClassWork03_Problem14_Rached.cpp -o p14
georgy@Georgy:~/EECE2140/Week2/Classwork$ ./p14
Enter positive integer: 2048
Digits: 4
```

Problem 15 First Multiple Above a Target

Read a positive divisor d and a nonnegative target t. Use a while loop to find and print the smallest positive multiple of d that is strictly greater than t.

1 Pseudocode

Initialize divisor, target, total
Cout and Cin for divisor and target
Make while loop to detect if divisor and target are below 0 and remake user put non-negative
Make another while loop, doing target module divisor to see if an even number is achieved (equation != 0 for even)
Cout total

2 C++ Source File

The .cpp file has been uploaded separately.

3 Output Screenshot

```
georgy@Georgy:~/EECE2140/Week2/Classwork$ g++ InClassWork03_Problem15_Rached.cpp -o p15
georgy@Georgy:~/EECE2140/Week2/Classwork$ ./p15
Enter divisor: 6
Enter target: 20
First multiple above target: 24
```

Problem 16 Prime Number Checker

Read an integer n greater than 1. Use a for loop and an if statement to decide whether it has a divisor between 2 and n - 1. Print “Prime” or “Not prime”. Test 2, 9, and 13.

1 Pseudocode

Input int n;
For I = 2 to n

If n % i == 0 then not prime
I++
If this never happens or n == 2
prime

2 C++ Source File

The .cpp file has been uploaded separately.

3 Output Screenshot

```
georgy@Georgy:~/EECE2140/Week2/Classwork$ g++ InClassWork03_Problem16_Rached.cpp -o p16
georgy@Georgy:~/EECE2140/Week2/Classwork$ ./p16
Enter n: 13
Prime
georgy@Georgy:~/EECE2140/Week2/Classwork$ ./p16
Enter n: 12
Not prime
```

Problem 17 Sensor Alarm Summary

Read exactly five temperature readings. For each reading, use nested if / else statements to classify it: below 0 is Cold; 0 through 35 inclusive is Safe; above 35 is Hot. At the end print the number of Cold and Hot readings.

1 Pseudocode

Initialize 5 temperature readings, hot, cold and safe
Cout and Cin each reading
If loop for each temp variable, if <0, cold, else -> another if statement if <=35 safe, else hot. Use nested here
Cout hot cold and safe variables

2 C++ Source File

The .cpp file has been uploaded separately.

3 Output Screenshot

```
georgy@Georgy:~/EECE2140/Week2/Classwork$ g++ InClassWork03_Problem17_Rached.cpp -o p17
georgy@Georgy:~/EECE2140/Week2/Classwork$ ./p17
Reading 1: -2
Reading 2: 10
Reading 3: 36
Reading 4: 35
Reading 5: 40
Cold: 1
Hot: 2
Safe: 2
```

Problem 18 Three-Attempt Access Code

Set a correct integer code in your program. Use a while loop to allow at most three attempts. After each incorrect attempt, use nested if / else logic to print either “Try again” or “Access locked”. Stop immediately when the code is correct.

1 Pseudocode

Int code = 4321
While input guess not equal code Try again until guess# = 3 or guess = code
If guess# = 3 lock
If input = code access granted

2 C++ Source File

The .cpp file has been uploaded separately.

3 Output Screenshot

```
georgy@Georgy:~/EECE2140/Week2/Classwork$ g++ InClassWork03_Problem18_Rached.cpp -o p18
georgy@Georgy:~/EECE2140/Week2/Classwork$ ./p18
Enter code: 1111
Try again
Enter code: 4321
Access Granted
georgy@Georgy:~/EECE2140/Week2/Classwork$ ./p18
Enter code: 1231
Try again
Enter code: 1324
Try again
Enter code: 1231
Try again
Access Locked
```

Problem 19 Traffic Light Controller

Read a positive number of time steps. At each step, read a pedestrian request (0 or 1) and the number of waiting cars. Use a for loop and nested if / else: if a pedestrian is waiting, print WALK; otherwise print GREEN if cars are waiting, or RED if there are no cars.

1 Pseudocode

Initialize steps, pedestrians, cars
Cout and cin for steps
Create for loops that stops when iteration exceeds steps
Ask for pedestrians and cars in for loop
Check for pedestrian with if, then nest another if to check if there is car, else for no car
Cout Walk, Green or Red depending on result

2 C++ Source File

The .cpp file has been uploaded separately.

3 Output Screenshot

```
georgy@Georgy:~/EECE2140/Week2/Classwork$ g++ InClassWork03_Problem19_Rached.cpp -o p19
georgy@Georgy:~/EECE2140/Week2/Classwork$ ./p19
Enter time steps: 2
Pedestrian request: 0
Waiting cars: 3
Step 1: GREEN
Pedestrian request: 1
Waiting cars: 4
Step 2: WALK
```

Problem 20 Elevator Trip Simulator

An elevator starts on floor 1. Read destination floors until the user enters 0. Valid floors are 1 through 10; reject other values without moving. For each valid request, use nested if / else to print UP, DOWN, or STAY, and accumulate the total number of floors traveled. Print the total when 0 is entered.

1 Pseudocode

Int currentfloor = 1
Int totalfloorstraveled = 0
Int destination = -1
While destination not 0
Input new val for destination
Move that many floors to it
Add that to the floors traveled count
If destination 0 then end and return floorstraveled

2 C++ Source File

The .cpp file has been uploaded separately.

3 Output Screenshot

```
georgy@Georgy:~/EECE2140/Week2/Classwork$ g++ InClassWork03_Problem20_Rached.cpp -o p20
georgy@Georgy:~/EECE2140/Week2/Classwork$ ./p20
Destination (0 to stop): 5
UP 4 floors
Destination (0 to stop): 3
DOWN 2 floors
Destination (0 to stop): 3
STAY
Destination (0 to stop): 0
Total floors traveled: 6
```